PhysT-Net: An Exogenous-Aware Spatio-Temporal Graph Transformer for Domain-Constrained Network Forecasting

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***Abstract*—Spatio-temporal forecasting is a core predictive task for complex graph-structured systems, which is important for operational decision-making in domains such as intelligent traffic routing and resource scheduling. However, existing methodologies frequently blur the temporal distinction between historical observations and future exogenous drivers, while struggling to natively satisfy strict physical operational constraints without relying on naive post-hoc clipping that may weaken gradient signals or create train-inference mismatch. To overcome these limitations, we propose PhysT-Net, a generalized exogenous-aware spatio-temporal graph Transformer framework for domain-constrained network forecasting. Specifically, the framework introduces Adaptive Gaussian-soft-windowed Patching (AGSP) for continuous temporal tokenization and a Relational Spatio-Temporal Manifold Fusion (R-STMF) adapter that utilizes a global token to efficiently decouple macro-trend and local-variation spatial dependencies. Furthermore, it incorporates KAN-enhanced Adaptive Modulation and Conditioning (K-AMC) for non-linear modulation of future exogenous factors, and a Physics-Informed Dynamic Capping Head (PI-DCH) that employs a differentiable barrier gate to promote constraint-aware decoding. The proposed formulation provides a reusable basis for evaluating domain-constrained forecasting systems under diverse operational regimes.**

***Index Terms*—Spatio-temporal forecasting, graph Transformers, exogenous variables, domain-constrained machine learning, network forecasting.**

I. Introduction

SPATIO-TEMPORAL forecasting predicts future states in graph-structured systems (e.g., transportation networks, power grids) where temporal dynamics and spatial interactions evolve jointly. Node-level trajectories are rarely self-driven; rather, they are shaped by historical conditions, future external drivers, and domain-specific feasibility constraints. As illustrated in Fig. 1, complex network signals comprise shared macro-trends driven by exogenous forcing and high-frequency local variations. Consequently, a practical forecasting model must learn continuous temporal representations, infer dynamic graph relations, and integrate future exogenous variables (ExG) without data leakage, all while maintaining operational compatibility.

A. The Forecasting Trilemma in Spatio-Temporal Networks

Despite recent architectural advancements, modern spatio-temporal forecasting frameworks remain hampered by a "Forecasting Trilemma"—an inherent trade-off between Model Capacity, Domain Consistency, and Spatio-Temporal Complexity.

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1) Model Capacity vs. High-Frequency Noise: Capacity denotes architectural expressive power. While modern Transformers excel at sequence modeling, they are highly sensitive to high-frequency stochastic noise in raw spatio-temporal data. Furthermore, rigid temporal patching exacerbates this by artificially splitting continuous physical regimes. As highlighted by recent trends in complexity-balanced sequence modeling, uniform representational capacity across an entire time horizon is computationally inefficient. The challenge is designing an architecture capable of capturing macro-trends while employing adaptive temporal allocation to filter node-specific noise.

2) Domain Consistency and Constraint-Aware Operations: Consistency mandates that a model's outputs adhere to the fundamental physical laws or operational limits inherent to the target domain. Purely data-driven architectures frequently generate unfeasible predictions—such as negative vehicular speeds or resource usage exceeding an operational upper bound. Such violations severely degrade operator trust in critical infrastructure. Drawing from emerging paradigms in safety-critical generation that utilize output-space guidance, a robust forecasting framework must enforce physical feasibility directly within its decoding pathway, moving beyond the limitations of reactive, post-hoc corrections.

3) Spatio-Temporal Complexity and Multi-Scale Dynamics: Complexity involves the management of dense spatial couplings across heterogeneous nodes. In physical networks, nodes interact via multiple simultaneous mechanisms: local topological dependencies (e.g., localized propagation effects, adjacent road segments) and global environmental fronts (e.g., system-wide contextual regimes). Existing models frequently conflate these multi-scale interactions into a singular spatial operator, thereby failing to distinguish localized perturbations from system-wide drivers.

To systematically resolve this trilemma, we propose PhysT-Net, an ExG-aware Spatio-Temporal Graph Transformer designed as a versatile framework for domain-constrained network forecasting. To accommodate the multi-scale and multi-modal nature of dynamic network signals, we introduce the following key innovations:

- **Continuous and Differentiable Patching**: We present the **Adaptive Gaussian-soft-windowed Patching (AGSP)** mechanism. By transforming discrete observations into learnable spectral-temporal densities, AGSP recovers continuous event dynamics and functions as an adaptive low-pass filter, effectively rejecting high-frequency noise.
- **Decoupled Spatio-Temporal Topology**: We introduce the **Relational Spatio-Temporal Manifold Fusion (R-STMF)** framework. R-STMF leverages a dual-branch

graph logic to decompose latent node dynamics into multi-scale spatial manifolds, mitigating the spatial conflation typical of standard distance-based graphs.
- **Piecewise Conditioning and Safe Capping**: We implement the **KAN-enhanced Adaptive Modulation and Conditioning (K-AMC)** module, utilizing Kolmogorov-Arnold spline edge functions for the non-linear modulation of graph-enhanced states via exogenous drivers. This is coupled with a **Physics-Informed Dynamic Capping Head (PI-DCH)** that embeds strict physical safety constraints and adherence to operational envelopes directly into the output space.

By explicitly partitioning the data flow into distinct computational pathways—historical context, relational reasoning, and exogenous conditioning—PhyST-Net establishes a rigorous forecasting contract. This interface-centric design prevents the entanglement of historical data and future drivers, ensuring the architecture can be seamlessly instantiated across diverse critical domains.

B. Contributions

The primary contributions of this work are summarized as follows:

- **A general ExG-aware graph-forecasting formulation.** We define network forecasting as a multi-modal task where historical states, exogenous variables, and domain constraints are processed through distinguishable pathways, thereby supporting broad architectural generalizability.
- **A continuous temporal tokenization module.** We introduce AGSP, which supersedes rigid patch boundaries with learnable Gaussian soft-windows, preserving event continuity and minimizing boundary artifacts in non-stationary sequences.
- **A modular graph adapter for mixed spatial mechanisms.** We develop R-STMF to disentangle multi-scale spatial dependencies prior to relational propagation, providing a modular contract that accommodates alternative graph predictors.
- **A future-ExG-conditioned decoding path.** We fuse K-AMC and PI-DCH to inject forecast-horizon drivers and structurally promote operational feasibility within the model, bridging latent representation learning with strict operational requirements.

The experimental section later instantiates this formulation on real-world forecasting tasks under diverse operational regimes.

II. Related Work

This section reviews representative research trajectories pertinent to PhyST-Net. Rather than chronicling models chronologically, we categorize prior literature based on the core modeling challenges inherent to ExG-aware spatio-temporal forecasting. A summary of the architectural taxonomies is provided in Table I.

A. Temporal Forecasting and Transformer Architectures

Classical statistical forecasting techniques, including autoregressive and state-space models, retain utility when predicting low-dimensional, approximately stationary processes. However, these foundational assumptions become restrictively fragile in graph-structured systems where node dynamics are highly non-linear and persistently coupled with external forcing. While deep recurrent and convolutional architectures improved non-linear sequence modeling via latent state tracking and local temporal filtering, long-horizon forecasting remains challenging due to error accumulation and vanishing gradients over extended time steps.

Transformer-based forecasters resolve long-range dependency constraints via self-attention, establishing themselves as the dominant paradigm for deep time-series analysis. Early breakthrough models, such as Informer [2], Autoformer [3], and FEDformer [4], focused on reducing the quadratic complexity of attention and decomposing trend-seasonal patterns. Recent variants, utilizing PatchTST-style patching and inverted-dimension attention, significantly reduce computational overhead while enhancing representation learning for multivariate sequences [1], [5], [6]. Nevertheless, temporal patching is predominantly implemented via rigid, fixed-length windows. In highly non-stationary sequences, physical regime transitions and rapid fluctuations rarely align with these artificial boundaries. PhyST-Net advances this trajectory by introducing AGSP, rendering temporal tokenization fully learnable and mathematically continuous.

B. Spatio-Temporal Graph Forecasting

Spatio-temporal graph neural networks (STGNNs) build upon foundational graph convolutions [7] and graph attention mechanisms [8] to capture node interactions. These methodologies are highly effective when cross-node dependencies dictate system behavior, such as in traffic corridors and power grids [9]–[11]. Furthermore, adaptive graph learning relaxes the restrictive assumption of a static, fully observable topology, allowing models to infer latent relations directly from data and dynamically adjust propagation patterns.

A critical limitation, however, is the widespread reliance on homogeneous propagation protocols across all latent states. In physical systems, spatial dependence is inherently multimodal; nodes interact through localized proximities, shared environmental influences, and overarching exogenous drivers. A singular adjacency matrix, even if learned dynamically, conflates these distinct mechanisms. R-STMF resolves this by decomposing temporal representations into high-frequency local variations and low-frequency macro-trends, applying targeted, sparse relational propagation to each manifold prior to fusion.

C. Exogenous-Variable-Aware Forecasting

Real-world forecasting is intrinsically dependent on exogenous (ExG) variables. In vertical domains, such as intelligent traffic routing—where models like ConFormer [12] leverage

TABLE I
TAXONOMY AND COMPARATIVE ANALYSIS OF SPATIO-TEMPORAL FORECASTING MODELS.

Model	Paradigm	Temporal Tokenization	Spatial Interaction	Exogenous Integration	Constraint Awareness
Autoformer / FEDformer	Transformer	Point-wise / Decomposition	None (Single-node)	Feature Concatenation	None
PatchTST / iTransformer	Transformer	Rigid Patches / Inverted	None (Single-node)	None / Covariate Tokens	None
STGCN / Graph WaveNet	ST-GNN	Convolutional / Recurrent	Homogeneous Graph Conv	Feature Concatenation	None
SDG / SDG-Net	ST-GNN	Convolutional	Latent Dynamic Graph	None	None
TimeXer / ConFormer	Hybrid	Rigid Patches	None / Cross-Attention	Variate Tokens (Cross-Attn)	None
GCGNet / ExoST	ST-GNN	Convolutional	Generative Graph Alignment	Joint Temporal-Channel	None
GeatFormer / PGNN-TFT	Physics-Inf.	Rigid Patches	Static Graph	None	Penalty / Post-hoc Clip
PhyST-Net (Ours)	**Hybrid**	**Continuous Gaussian (AGSP)**	**Decoupled Manifold (R-STMF)**	**KAN Conditioning (K-AMC)**	**Diff. Head (PI-DCH)**

external factors like accidents and regulations—and resource-constrained operational forecasting, ExG features are critical for predictive resilience.

Recent foundational architectures have begun systematically targeting ExG fusion to address inconsistent covariate effects [13]. In time-series modeling, TimeXer [14] empowers Transformers by mapping external factors into variate tokens, enabling cross-attention between endogenous patches and exogenous drivers. Concurrently, DAG [15] introduces dual correlation networks to capture channel-wise dependencies, while in the spatio-temporal graph domain, models like GCGNet [16] employ generative frameworks to align predicted correlations with joint temporal-channel graphs driven by external factors.

While these mechanisms significantly improve accuracy under ideal conditions, they frequently blur the temporal distinction between historical observations and future drivers. Specifically, historical ExG elucidates recent state trajectories, whereas future ExG provides horizon-level modulation. Merging both into a singular feature tensor exacerbates temporal imbalance effects [13] and complicates data availability reasoning at inference time. Consequently, PhyST-Net structurally decouples historical ExG from future ExG. K-AMC leverages the future-ExG pathway to generate latent modulation parameters, enabling forecast-horizon drivers to dynamically condition graph-enhanced states without being erroneously treated as historical priors.

D. Interpretable Conditional Modeling with KANs

Interpretability is paramount in operational forecasting, as dispatchers require insight into the primary drivers shaping a prediction. While attention weights and post-hoc attribution methods offer diagnostic utility, they often fail to reflect the model's actual computational pathways. Kolmogorov-Arnold Networks (KAN) [19] represent a paradigm shift from traditional perceptrons by modeling transformations via learnable, univariate spline-based edge functions [20]–[23].

By structuring a conditioning network as an additive sum of feature-wise edge functions, the influence of individual input variables can be analytically inspected through learned basis coefficients and derivatives. PhyST-Net capitalizes on this property within the K-AMC module: future ExG features are mapped to scale and shift parameters via spline functions. While not establishing strict causal identifiability, this architectural alignment provides a transparent, inspectable explanation mechanism far superior to generic dense concatenation.

E. Physics-Informed and Constraint-Aware Forecasting

Physics-informed learning embeds essential domain knowledge into neural architectures via structural priors, penalty losses, or projection operators. In network forecasting, enforcing such constraints is vital, as system targets are strictly bounded by operational capacities. Unfeasible predictions, regardless of aggregate error metrics, can trigger catastrophic downstream decisions.

Contemporary constraint-aware frameworks predominantly rely on physics-guided loss regularization [24], [25] or reactive post-hoc clipping [26], [27]. However, soft penalties do not guarantee compliance, and post-hoc clipping creates a structural disconnect between the trained model and inference behavior. Aligning with the philosophy of structural physics induction [28], the PI-DCH layer internalizes feasibility operations directly within the forecasting head. By utilizing smooth barrier functions to shape predictions toward a dynamic capacity envelope prior to a final non-linear projection, the constraint pathway becomes an explicit, differentiable component of the computational graph.

F. Position of This Work

PhyST-Net synthesizes advancements in Transformer forecasting, adaptive graph learning, and interpretable exogenous conditioning. Its distinguishing factor lies in the rigorous organization of the forecasting interface. Rather than presenting a bespoke, single-domain architecture, this paper formulates a generalized ExG-aware graph prediction framework, subsequently instantiating physically meaningful, domain-specific modules. This methodology positions PhyST-Net as a reusable framework for domain-constrained network forecasting.

III. PROBLEM DEFINITION

We formulate the task as exogenous-variable-aware (ExG-aware) spatio-temporal graph forecasting. This formulation aims to predict future states of a networked system by integrating historical observations, graph-structured dependencies, and external driving factors.

A. Graph Representation and Spatial Dependencies

Let $G = (V, E, A)$ denote a spatio-temporal graph, where $V = \{v_1, \ldots, v_N\}$ is the set of N nodes representing physical or logical entities (e.g., wind turbines in a farm or sensors in a city). E is the set of edges representing connections, and $A \in \mathbb{R}^{N\times N}$ is the adjacency matrix encoding spatial relations. In practice, A can be a physical adjacency matrix, a distance-based kernel, or a learned correlation matrix. To maintain flexibility across different domains, we further define M_G as a set of graph metadata, containing auxiliary information such as node coordinates, static attributes, or environmental descriptors that are not captured by the topology A alone.

B. Temporal Tensors and Exogenous Variables

At a given forecasting origin t, we define the historical target tensor $\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T\times N\times D_x}$, where T is the look-back window and D_x is the dimension of the target state at each node. Our goal is to predict the future target tensor $\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H\times N\times D_y}$ over a forecast horizon H.

A critical challenge in modern forecasting is the integration of exogenous variables (ExG), which we categorize into two distinct types based on their temporal availability:

- **Historical ExG** ($\mathbf{E}^h_{t-T+1:t} \in \mathbb{R}^{T\times D_h}$): These variables represent external conditions (e.g., historical ambient temperature or humidity) that were observed alongside the target state history. They provide contextual information to help the model identify the latent state of the system during the look-back period.
- **Future ExG** ($\mathbf{E}^f_{t+1:t+H} \in \mathbb{R}^{H\times D_f}$): These represent known or forecastable future drivers (e.g., weather forecasts or scheduled operational constraints) that are available for the duration of the forecast horizon. Unlike historical variables, future ExG serves as a conditional driver for the unknown future trajectory.

C. Operational Forecasting Assumptions

To ensure the model is applicable to real-world operational settings, we enforce a strict non-leakage constraint. At inference time t, the model f_θ is only permitted to access information available up to t for target states and historical ExG, and only those future ExG variables that are genuinely known or forecast by external sources (e.g., Numerical Weather Prediction models). This ensures that $\mathbf{E}^f$ does not contain any "future-peeking" information that would be unavailable in a deployed environment.

D. General Optimization Objective

To simplify the notation, we encapsulate all input drivers available at time t into a single input tuple $\mathcal{X}_t$:

$$\mathcal{X}_t \triangleq \big(\mathbf{X}_{t-T+1:t}, \mathbf{E}^h_{t-T+1:t}, \mathbf{E}^f_{t+1:t+H}, A, M_G\big). \quad (1)$$

The forecasting task is defined by a parameterized model f_θ which maps $\mathcal{X}_t$ to the predicted horizon $\hat{\mathbf{Y}}_{t+1:t+H}$. For domains where physical laws or operational limits must be satisfied, the model can be decomposed into an unconstrained spatio-temporal backbone $\mathcal{H}_\theta$ and a constraint operator $\mathcal{C}_\Omega$ parameterized by domain knowledge Ω:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{C}_\Omega\big(\mathcal{H}_\theta(\mathcal{X}_t)\big). \quad (2)$$

The operator $\mathcal{C}_\Omega$ ensures that the final output adheres to feasibility requirements such as non-negativity or capacity limits.

Given a training dataset $\mathcal{D}$, the model parameters θ are optimized by minimizing the Mean Absolute Error (MAE) between the predictions and the ground truth:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|\cdot H\cdot N} \sum_{(\mathcal{X},\mathbf{Y})\in\mathcal{D}} \sum_{h=1}^{H}\sum_{n=1}^{N} |f_\theta(\mathcal{X})_{h,n} - \mathbf{Y}_{h,n}|, \quad (3)$$

where the L_1 loss provides robustness against outliers in the spatio-temporal data. This objective provides a general training framework that can be easily extended with additional task-specific regularizers in later stages of the methodology.

IV. Method

PhyST-Net is an ExG-aware spatio-temporal graph Transformer designed for network-structured node forecasting with known external drivers and domain constraints. Given the inputs defined in Section III, the model follows the pipeline in Fig. 1. Historical target states and historical ExG are first embedded into node-level temporal representations. AGSP converts the temporal history into continuous tokens. A Transformer backbone learns temporal dependencies. R-STMF serves as a graph adapter that maps temporal node states to graph-enhanced node states. K-AMC injects future ExG into the latent representation. PI-DCH finally decodes the conditioned states into feasible forecasts respecting domain-specific bounds.

A. Overall Framework

For a mini-batch, let $\mathbf{X} \in \mathbb{R}^{B\times T\times N\times D_x}$ denote historical target states, $\mathbf{E}^h \in \mathbb{R}^{B\times T\times D_h}$ denote historical ExG, and $\mathbf{E}^f \in \mathbb{R}^{B\times H\times D_f}$ denote future ExG. The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{B\times H\times N\times D_y}$. The computation can be written as

$$Z = \Phi_{patch}(\mathbf{X}, \mathbf{E}^h, M_G), \quad (4)$$

$$H_T = \Phi_{temp}(Z), \quad H_G = \Phi_{graph}(H_T, A, M_G), \quad (5)$$

$$H_F = \Phi_{exg}(H_G, \mathbf{E}^f), \quad \hat{\mathbf{Y}} = \Phi_{head}(H_F, \mathbf{E}^f, \Omega), \quad (6)$$

where Φ_{patch} is AGSP, Φ_{temp} is the temporal Transformer encoder, Φ_{graph} is the graph adapter, Φ_{exg} is future-ExG conditioning (K-AMC), and Φ_{head} is the physics-informed prediction head (PI-DCH). This decomposition emphasizes that the architectural backbone is defined on generic graph-structured sequences, while domain-specific knowledge is encapsulated in the input interface and the constraint head.

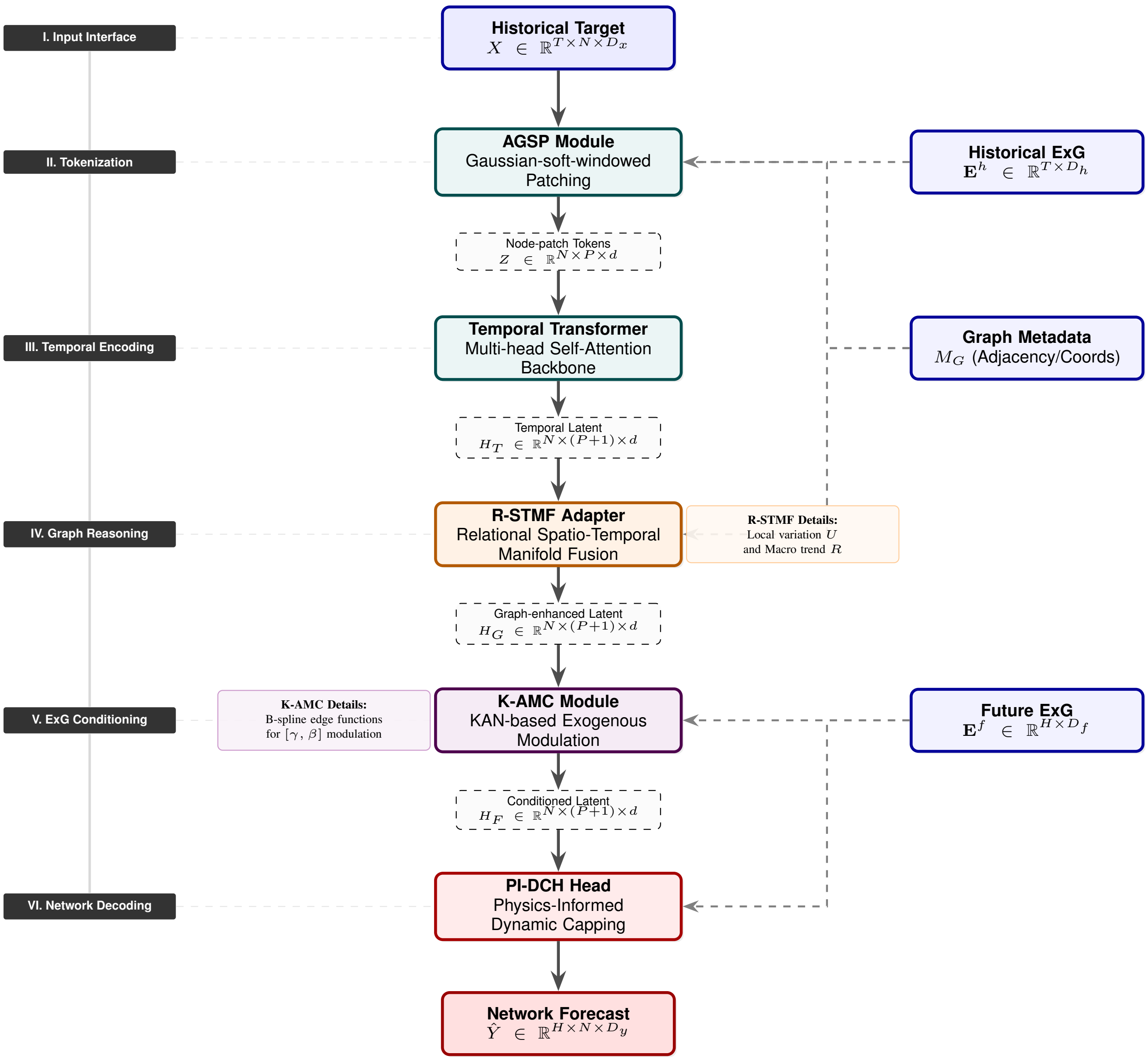

Fig. 1. Overall architecture of PhyST-Net. The framework operates as an exogenous-aware vertical pipeline: (I) Multi-modal input interface; (II) AGSP continuous tokenization; (III) Transformer sequence encoding; (IV) R-STMF relational manifold fusion; (V) K-AMC spline-based conditioning; and (VI) PI-DCH domain-constrained forecasting. All external data (ExG and Metadata) and technical details are organized in peripheral tracks to ensure a clear and interpretable data flow.

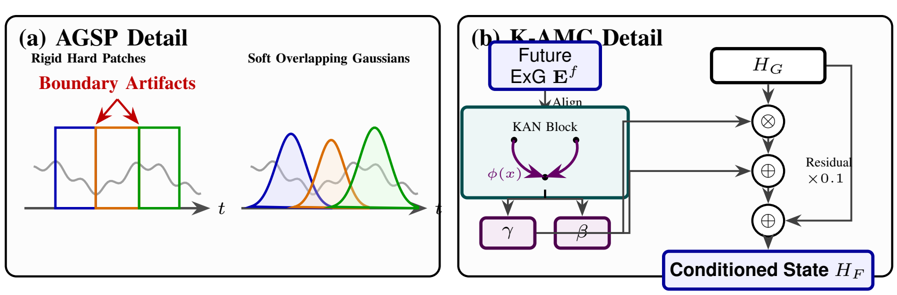

Fig. 2. Internal mechanics of the coupled modules. (a) AGSP replaces discrete sequence chunking with continuous, differentiable density windows, reducing boundary artifacts. (b) K-AMC leverages Kolmogorov-Arnold Networks (KAN) to map future exogenous features into dynamic scale (γ) and shift (β) parameters for conditional modulation.

B. Adaptive Gaussian-soft-windowed Patching

Transformer-based forecasters conventionally mitigate sequence length by aggregating observations into discrete temporal patches. While computationally convenient, fixed non-overlapping patches inevitably induce boundary artifacts when the underlying physical signal transverses a patch boundary. Aligning with emerging principles in complexity-balanced sequence modeling—where representational capacity is dynamically distributed proportional to signal variation—AGSP supersedes rigid patch assignments with learnable Gaussian soft-windows. This differentiable formulation empowers the architecture to autonomously allocate dense receptive fields to highly transient events and broader, smoother fields to stable operational regimes.

For node n and temporal window i, the unnormalized weight at time index $\tau \in \{1, \dots, T\}$ is

$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right), \tag{7}$$

where μ_i and σ_i are learnable center and bandwidth parameters. To ensure numerical stability and valid window properties, we parameterize the bandwidth as $\sigma_i = \mathrm{softplus}(\rho_i) + \delta$, where ρ_i is the learnable parameter. The centers μ_i are initialized to uniform grid locations across the historical window to encourage an ordered temporal partition. The weights are normalized across P windows:

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^{P} W_j(\tau) + \epsilon}. \tag{8}$$

Let $\bar{x}_n(\tau)$ denote the embedded historical state. The token for node n and patch i is

$$z_{n,i} = \psi\left(\sum_{\tau=1}^{T} \tilde{W}_i(\tau)\bar{x}_n(\tau)\right) + e_n + p_i, \tag{9}$$

where $\psi(\cdot)$ is a linear projection, e_n is a node embedding, and p_i is a temporal patch embedding. This formulation allows the model to optimize the temporal receptive fields directly from the forecast loss.

C. Temporal Representation Learning

Adopting the global token philosophy [14] to facilitate efficient long-range context fusion, the token tensor Z is concatenated with a learnable global token $z_{global} \in \mathbb{R}^{B\times N\times 1\times d}$ for each node. This global token serves as a sequence-level memory bottleneck, aggregating the macroscopic temporal representation (e.g., shared external regimes or system-wide traffic trends) across all local patches.

We denote the temporal encoding backbone as $\mathrm{STEnc}(\cdot)$. In PhyST-Net, rather than a naive standard Transformer, STEnc incorporates a targeted cross-attention mechanism. Specifically, within each encoder layer, the augmented node-wise token sequence $[Z \parallel z_{global}]$ first undergoes standard multi-head self-attention. Subsequently, to enrich the macro-trend representation with system-wide context, the global token uniquely cross-attends to the embedded historical exogenous features (e.g., SCADA and time-mark metadata), acting as an information sink:

$$z'_{global} = z_{global} + \mathrm{CrossAttention}(z_{global}, \mathbf{E}^h, \mathbf{E}^h). \tag{10}$$

This contextually enriched sequence is then processed through the feed-forward networks, yielding the final temporal representation:

$$H_T = \mathrm{STEnc}([Z \parallel z_{global}]), \quad H_T \in \mathbb{R}^{B\times N\times (P+1)\times d}. \tag{11}$$

Crucially, rather than discarding the global token post-encoding, PhyST-Net structurally leverages it to separate the node's state into two distinct manifolds: a local-variation patch state $U \triangleq H_{T,1:P}$ and a macro-trend global state $R \triangleq H_{T,P+1}$.

D. Relational Spatio-Temporal Manifold Fusion (R-STMF)

R-STMF serves as the graph adapter, specifically designed to capture the multi-modal spatial dependencies inherent in physical networks. Unlike conventional graph operators that treat node features as monolithic vectors, R-STMF recognizes that spatial interactions occur across different temporal scales within the latent space. To address this, we perform a Latent Patch Manifold Decomposition on the temporal representation $H_T \in \mathbb{R}^{B\times N\times P\times d}$.

Specifically, for each node n, we treat the sequence of P latent patches as a low-resolution temporal manifold. We apply a 1D depth-wise convolution along the patch dimension to extract the seasonal (periodic/oscillatory) component H_S, while the residual represents the trend (drifting/low-frequency) component H_R:

$$H_S = \mathrm{GELU}(\mathrm{Conv1d}(H_T)), \tag{12}$$

$$H_R = H_T - H_S. \tag{13}$$

This decomposition ensures that the model can independently learn how macro-scale trends and micro-scale seasonal fluctuations propagate through the graph.

Distance-Masked Relational Pruning: For each manifold $c \in \{S, R\}$, we compute relational attention scores across nodes. To enforce physical realism and reduce computational overhead, we introduce a **Distance-Masked Top-K Pruning** strategy. Given a physical distance matrix or prior topology, we define a binary mask $\mathcal{M}_{dist}$, where $\mathcal{M}_{ij} = 1$ only if node j is within a feasible interaction radius of node i. The masked attention score S_c is:

$$S_{c,ij} = \begin{cases} \frac{Q_c(K_c)^\top}{\sqrt{d_m}} & \text{if } \mathcal{M}_{ij} = 1 \\ -\infty & \text{otherwise} \end{cases}. \tag{14}$$

Subsequently, we perform Top-K selection on the remaining scores to ensure a sparse and robust graph routing:

$$A_c = \mathrm{Softmax}\left(\mathrm{TopK}(S_c, K_c)\right). \tag{15}$$

Finally, the graph-enhanced seasonal and trend manifolds are recombined using a learnable gated fusion mechanism:

$$g = \sigma(\mathrm{Linear}([H_{G,S} \parallel H_{G,R}])), \tag{16}$$

$$H_G = g \odot H_{G,S} + (1-g) \odot H_{G,R}, \tag{17}$$

where $H_{G,c}$ denotes the manifold c after spatial propagation via A_c. This dual-branch architecture allows PhyST-Net to model complex spatio-temporal interactions without conflating multi-resolution physical mechanisms.

E. Future-ExG Conditioning (K-AMC)

We refer to this module as K-AMC, short for KAN-enhanced Adaptive Modulation and Conditioning. It serves as a generalized interface for known systemic drivers, injecting future exogenous drivers $\mathbf{E}^f \in \mathbb{R}^{B\times H\times D_f}$ into the latent representations. Aligning with the TimeXer philosophy [14], the macro-trend global token acts as the primary bridge to cross-attend and ingest external factors. We define an alignment operator $\Gamma : \mathbb{R}^{H\times D_f} \to \mathbb{R}^{(P+1)\times d_{exg}}$ that maps the future horizon into variate tokens corresponding to the global and patch states.

The aligned representation $\bar{\mathbf{E}}^f$ is then passed through a Kolmogorov-Arnold Network (KAN) block. Unlike standard MLPs, KANs parameterize transformations using learnable univariate B-spline functions on the edges. For an input feature x, the edge function is defined as:

$$\phi(x) = w_b\,\mathrm{SiLU}(x) + \sum_{i=1}^{k} c_i B_i(x), \tag{18}$$

where $B_i(x)$ are B-spline basis functions. The analytical derivative is:

$$\frac{\partial \phi(x)}{\partial x} = w_b\,\mathrm{SiLU}'(x) + \sum_{i=1}^{k} c_i B_i'(x). \tag{19}$$

By aggregating these edge functions, the KAN block produces modulation parameters:

$$[\gamma, \beta] = \mathrm{KAN}(\bar{\mathbf{E}}^f), \quad \gamma, \beta \in \mathbb{R}^{B\times N\times (P+1)\times d}. \tag{20}$$

The conditioned latent state is then obtained via element-wise scale and shift:

$$H_F = H_G + \lambda\left(\gamma \odot H_G + \beta - H_G\right). \tag{21}$$

F. Physics-Informed Dynamic Capping Head (PI-DCH)

PI-DCH instantiates the constraint operator $\mathcal{C}_\Omega$. In mission-critical environments such as smart grids, safety-critical traffic routing, and industrial control systems, deploying predictive models mandates rigorous output-space guidance to preclude physically unfeasible trajectories. While hard projection operators (e.g., clipping) guarantee bounded outputs, they may create a mismatch between training and inference, and can severely weaken gradient signals when used naively, hampering the learning process.

PI-DCH circumvents this by systematically steering latent predictions toward a dynamic feasible envelope C_{dyn} via a differentiable barrier before applying the strict physical boundary. Specifically, a linear decoder first generates an unconstrained predictive tensor $\tilde{Y} = \mathrm{Linear}(H_F)$. Concurrently, a learned barrier gate $u = \sigma(g(H_F))$ modulates this prediction smoothly:

$$\hat{Y}_{soft} = \tilde{Y} + u \odot (C_{dyn} - \tilde{Y}). \tag{22}$$

Crucially, during the training phase, this differentiable soft-capped tensor $\hat{Y}_{soft}$ is utilized to compute the forecasting loss ($\mathcal{L}_{mae}$) and the penalty ($\mathcal{L}_{penalty}$). This guarantees that gradients backpropagate natively through the constraint-shaping logic, mitigating the train-inference mismatch of naive clipping.

The definitive feasible forecast $\hat{\mathbf{Y}}$ is subsequently derived via a non-linear hard projection:

$$\hat{\mathbf{Y}} = \mathrm{clip}(\hat{Y}_{soft}, 0, C_{dyn}). \tag{23}$$

This physics-informed projection is stringently enforced during inference, enforcing bounded feasibility with respect to the specified envelope.

G. Computational Complexity Analysis

A core motivation of PhyST-Net is mitigating the severe computational bottlenecks of dense spatio-temporal Transformers, which conventionally exhibit a time complexity of $\mathcal{O}(N \cdot T^2 + T \cdot N^2)$ for sequence length T and node count N. PhyST-Net structurally resolves this through its decoupled topology.

In the temporal pathway, the AGSP module compresses the raw T-length sequence into P continuous tokens (where $P \ll T$). Consequently, the temporal Transformer encoder's self-attention complexity is reduced from $\mathcal{O}(N \cdot T^2)$ to $\mathcal{O}(N \cdot P^2)$.

In the spatial pathway, standard full-graph attention requires $\mathcal{O}(P \cdot N^2)$ operations. The R-STMF module introduces a Top-K sparse masking strategy (Eq. 15), restricting message passing to a parameterized neighborhood size K. This bounds the spatial complexity to $\mathcal{O}(P \cdot K \cdot N)$. Thus, the overall self-attention complexity of the PhyST-Net backbone is decoupled and bounded by $\mathcal{O}(N \cdot P^2 + P \cdot K \cdot N)$, rendering the architecture highly scalable for large physical networks and extended forecasting horizons without sacrificing expressive capability.

H. Optimization Objective

To bridge the gap between purely data-driven forecasting and physical feasibility, we minimize a multi-objective loss $\mathcal{L} = \mathcal{L}_{mae} + \lambda_\Omega \mathcal{L}_{penalty}$. The penalty term $\mathcal{L}_{penalty}$ structurally regularizes the model to internalize the operational set Ω during backpropagation. Utilizing the positive-part operator $[x]^+ \triangleq \max(0, x)$, we decouple the penalty into two physically meaningful components: a non-negativity constraint and a dynamic capacity limit constraint:

$$\mathcal{L}_{penalty} = \mathcal{L}_{neg} + \mathcal{L}_{cap}, \tag{24}$$

$$\mathcal{L}_{neg} = \frac{1}{|\mathcal{D}|HN} \sum_{\mathcal{D}} \sum_{n,h} [-\hat{Y}_{soft,n,h}]^+, \tag{25}$$

$$\mathcal{L}_{cap} = \frac{1}{|\mathcal{D}|HN} \sum_{\mathcal{D}} \sum_{n,h} [\hat{Y}_{soft,n,h} - C_{dyn,n,h}]^+. \tag{26}$$

This decoupled formulation ensures the model learns to respect physical boundaries natively, thereby reducing the reliance on reactive clipping at inference time. The overall training logic is summarized in Algorithm 1.

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Algorithm 1 PhyST-Net Forward Pass

Require: Input tuple $\mathcal{X}_t = (\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G)$, Domain knowledge Ω.
Ensure: Feasible network forecast $\hat{\mathbf{Y}}$.

{*Phase I: AGSP Continuous Tokenization*}
Initialize learnable Gaussian parameters μ_i, σ_i for P tokens using metadata M_G.
for each token $i \in \{1, \ldots, P\}$ **do**
  Compute soft-window weights $W_i(\tau) = \exp\left(-\frac{(\tau-\mu_i)^2}{2\sigma_i^2+\epsilon}\right)$.
  Aggregate $\mathbf{X}$ and $\mathbf{E}^h$ via $W_i(\tau)$ to form continuous token $Z_{:,i,:}$.
end for
{*Phase II: Temporal Encoding ($\mathcal{H}_\theta$ part 1)*}
Process Z via Multi-Head Self-Attention to capture temporal dependencies $\rightarrow H_T$.
{*Phase III: R-STMF Graph Adaptation ($\mathcal{H}_\theta$ part 2)*}
Decompose H_T into macro-trend R and local-variation U manifolds.
Apply Top-K sparse masking to A based on spatial similarities.
Propagate R and U over the masked graph and fuse $\rightarrow H_G$.
{*Phase IV: K-AMC Exogenous Modulation ($\mathcal{H}_\theta$ part 3)*}
Align horizon-wise $\mathbf{E}^f$ to patch-wise representation $\bar{\mathbf{E}}^f$.
Compute B-spline edge functions via KAN block to generate $[\gamma, \beta]$.
Modulate latent states: $H_F \leftarrow H_G + \lambda(\gamma \odot H_G + \beta - H_G)$.
{*Phase V: PI-DCH Constraint Decoding ($\mathcal{C}_\Omega$)*}
Decode unconstrained prediction $\tilde{Y} = \mathrm{Linear}(H_F)$.
Compute barrier gate $u = \sigma(g(H_F))$ to steer toward dynamic envelope $C_{dyn} \in \Omega$.
Shape prediction softly: $\hat{Y}_{soft} \leftarrow \tilde{Y} + u \odot (C_{dyn} - \tilde{Y})$.
Project to feasible set: $\hat{\mathbf{Y}} \leftarrow \mathrm{clip}(\hat{Y}_{soft}, 0, C_{dyn})$.
return $\hat{\mathbf{Y}}$.

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